

Cellulose Hydrogels by Reversible Ion-Exchange as Flexible Pressure Sensors

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Inspired by biological camouflage strategy according to the surrounding environment, a facile method to fabricate state-switchable cellulose hydrogels (fluid, brittle, and rigid hydrogel) driven by $\text{Ca}^{2+}/\text{Zn}^{2+}$ ion exchange is reported. The high content of Ca^{2+} ions leads to the formation of a rigid hydrogel (Gel-H- Ca^{2+}) with a great compressive strength (≈ 2.2 MPa) through coordination cross-linking network, while Zn^{2+} ions give the cellulose (Sol-L- Zn^{2+}) a fluid state by eliminating the cellulose intermolecular connections, making it possible to obtain injectable and healable hydrogels. Gel-H- Ca^{2+} is highly oriented and anisotropic, and such flexible hydrogel can effectively monitor slight bending and pressure changes of fingers. This cellulose hydrogel will be a promising candidate for bionic and multifunctional sensors.

metal ions and carboxyl groups or amino groups.^[6] For example, the sodium alginate (SA) hydrogel can precisely change into designed structures and recover their initial shapes by adjusting the contents of biocompatible ions ($\text{Na}^+/\text{Ca}^{2+}$).^[7] Complex 3D or even 4D folding of Ca-SA/polyacrylamide hydrogels were realized by controlling cross-linking of Fe^{3+} and Ca^{2+} ions locally.^[8] Ascribing to the strong binding between metal ions and organic groups, the sol-gel transition often depends on external conditions, such as temperature, pH value, and redox reactions.^[9] Noticeably, the effective integration of ions into hydrogel network is still a profound challenge.

1. Introduction

Metal ions are involved in biochemical reactions in many organisms and tissues, allowing them to form a variety of states.^[1] For example, the combination and separation between Ca^{2+} ions and troponin in sarcoplasm could realize muscle expansion and contraction.^[2] This ion-driven mechanism on synthetic materials is highly desirable.^[3] Inspired by this mechanism, various types of stretchable, transparent, and compressible ionic hydrogels have been developed and applied as artificial muscles, wearable sensors, skin, and touch panels.^[4]

A general way to introduce ions into hydrogels is by soaking the pre-prepared gel in various aqueous solutions with ionic compounds. Prior studies demonstrated that ionic hydrogels exhibit a remarkable anti-freezing property and can maintain high stretchability and good electrical conductivity at subzero temperatures.^[5] Moreover, ionic hydrogels can achieve shape and soft-stiff transformation by the dynamic cross-linking of

Natural polymer-based hydrogels have received tremendous considerations due to its inherent biocompatibility, biodegradability, and nontoxicity.^[10] As the most abundant polysaccharide in nature, plant cellulose is regarded as a promising starting material for hydrogel fabrication. In this context, cellulose ionic hydrogels have been recently reported as strain sensor,^[11] soft actuator,^[12] and tissue culture substrate.^[13] In particular, based on the unique semi-rigid chain^[14] and high-density hydrogen bonding structure,^[15] cellulose hydrogels with highly ordered structure can be prepared by pre-stretching method for improving mechanical strength and optics sensing.^[14b,16] Such properties make cellulose hydrogel a promising candidate to fabricate smart materials.^[17]

Very recently, inorganic salt solutions ($\text{ZnCl}_2/\text{CaCl}_2$) have been proven to be a green solvent to dissolve and cross-link cellulose.^[18] Zn^{2+} and Ca^{2+} ions were entrapped into hydrogel matrix during the dissolution process. In this study, we find that different cellulose states (fluid, brittle, and rigid hydrogel) can be reversibly switched by $\text{Ca}^{2+}/\text{Zn}^{2+}$ ion exchange at room temperature. Moreover, the ion exchange process has a great impact upon the anisotropy within the cellulose hydrogels. A high content of Ca^{2+} ions leads to alignment of cellulose chains, while Zn^{2+} ions can eliminate cross-linking sites and reduce the orientation. Therefore, cellulose ionic hydrogels cross-linked by Ca^{2+} ions would exhibit sensitive optical and electric responses under a small strain.

2. Results and Discussion

2.1. Preparation of Hydrogels

Cotton cellulose was used as the starting material and dissolved in the $\text{ZnCl}_2/\text{CaCl}_2$ solvent to prepare the cellulose hydrogel.^[18a]

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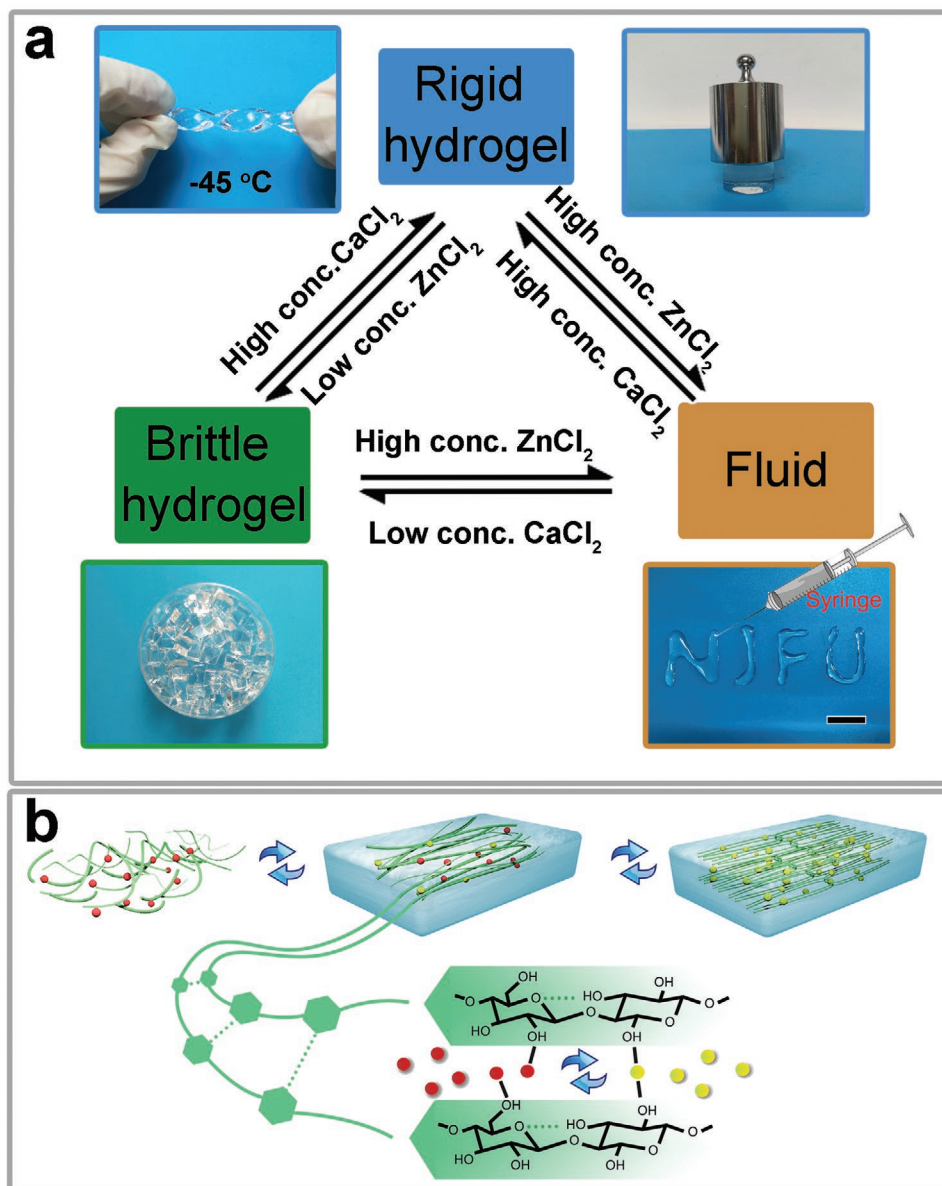


Figure 1. Schematic illustration of a) the reversible transition of cellulose hydrogels based on the ion exchange and the corresponding performance (scale bar = 2 cm) and b) the proposed mechanism.

At room temperature, the hydrogel was soaked in a high concentration of CaCl_2 solution (5.77 mol L^{-1}) for 2 h and repeated three times. In this process, most of Zn^{2+} ions in the hydrogel can be replaced by Ca^{2+} ions. The resulting hydrogels (Gel-H-Ca^{2+}) exhibited a rigid property and could bear the pressure of 100 g counterweight without breaking (Figure 1a). At an extremely low temperature condition (-45°C), the hydrogel remains elastic after repeated twisting. Interestingly when the hydrogel was soaked in a 4.21 mol L^{-1} ZnCl_2 solution for 12 h, the rigid hydrogel transformed into a brittle gel (Gel-L-Ca^{2+}) due to the loss of Ca^{2+} ions. The Gel-L-Ca^{2+} was crushed into pieces under the same pressure of the counterweight (100 g). From the compress stress-strain test (Figure S1, Supporting Information), the fracture stress of Gel-H-Ca^{2+} is up to 2.2 MPa, significantly higher than that of Gel-L-Ca^{2+} (0.7 MPa).

However, if Gel-L-Ca^{2+} was treated with a high concentration of ZnCl_2 solution (7.98 mol L^{-1}), the hydrogel will gradually turn into the fluid state (Sol-H-Zn^{2+}). Furthermore, by soaking in CaCl_2 solution with different concentrations, the Sol-H-Zn^{2+} can successfully return to the previous two gel states. Due to the viscous fluid properties, the Sol-H-Zn^{2+} is injectable and it could transfer into solid state after Ca^{2+} ion exchange. As shown in the bottom of Figure 1a, a 20 mL syringe was used to extrude Sol-H-Zn^{2+} via an 18-gauge needle, and the hydrogel letters solidified within a few seconds when CaCl_2 solution (5.77 mol L^{-1}) was dripped on the fluid letters. These letters have enough strength without breaking when lifted (the whole process of fluid writing and solidification were demonstrated in the Video S1, Supporting Information).

The reversible conversion between different states was realized only through ion exchange at room temperature. The

multi-state conversion mechanism was proposed and vividly depicted in Figure 1b. In a high concentration of ZnCl_2 , the Zn^{2+} ions were coordinated with hydroxyl groups in cellulose molecules.^[19] Because the intermolecular connection of cellulose was absent, the hydrogel presented a fluid state. When Ca^{2+} ions were introduced, they would compete with Zn^{2+} ions in the hydrogel and occupy the coordination sites with hydroxyl groups and connect the adjacent cellulose molecules, forming a cross-linking structure. As the concentration of CaCl_2 increased, the cross-linking density of Ca^{2+} ions and cellulose hydroxyl groups increased, thereby giving the hydrogel enhanced mechanical properties. Although the coordination types of Ca^{2+} and Zn^{2+} ions are different, the weak binding of the two metal ions and the hydroxyl group (proved by density functional theory (DFT) calculation in Figure 2a and Table S1, Supporting Information) endowed the cellulose hydrogel with reversible multi-state convention by ion exchange.

2.2. Characterization of Hydrogels

DFT modeling helps us further explore and determine the fundamental driving force of the reversible state transition via ion exchange. For both considered metals, the most energetically favorable configurations were corresponding with coordination by oxygen from sugar ring and hydroxyl group (Figure 2a-i,ii) and another with coordination by two hydroxyl group (Figure 2a-iii,iv). Based on the calculated binding energies and activation energy (DFT calculation in ESI), the coordination of metal atom by two hydroxyl groups (Figure 2a-iii,iv) was the most energetically favorable. Moreover, the energies for the bonding of Zn and Ca were -40.13 and -58.81 kcal mol⁻¹, respectively. Additionally, the activation energies of the metal-cellulose bonds were further calculated. These energies are defined as energy cost of displacement of metal-water complex away from cellulose at 1 Å. The values of activation energies are 59.96 and 64.50 kcal mol⁻¹ for zinc and calcium, respectively (note: because considering the removal of the ions, the activation energy is positive). In contrast to the binding energies that includes the energy of extraction of ions from water, the activation energy describes only the energy of interactions between metal and cellulose, thereby diminishing the difference between values for Zn and Ca. Therefore, it is reasonable to propose that the ion exchange process between different ions could be driven by ionic concentration.

The microstructures of the state-transition cellulose were further examined by a FT-IR analysis (Figure 2b). For comparison, the regenerated cellulose without ions was set as the control sample. It was prepared by soaking the Gel-L- Ca^{2+} in deionized water for 12 h and repeated until almost no ions in solution were detected. For the control sample, the absorption bands at 997, 1023, 1047, 1067 cm⁻¹ were observed, corresponding to the C–O valence vibration from C6–O6H and C3–O3H.^[18b] When Zn^{2+} or Ca^{2+} ions were introduced into cellulose, the intensity of band at 1047 cm⁻¹ significantly decreased, and the bands at 1023 cm⁻¹ disappeared. Alternatively, a new absorption peak at 1016 cm⁻¹ appeared, which could be attributed to the presence of OH–Zn and OH–Ca interactions. On the other hand, the peaks dominated from 3200 to 3400 cm⁻¹, corresponding

to the intermolecular hydrogen bonds (O6–HO3) and intramolecular hydrogen bonds (O3–HO5),^[20] shifted toward higher wavenumber, also confirming the formation of coordination structures.

The dynamic rheological tests were performed to reflect the strength and dynamic viscoelastic behavior of the network structure of hydrogels. The storage modulus (G') of all the hydrogels was several times greater than the loss modulus (G'') (Figure 2c), indicating a substantial elastic. At ≈ 1 rad s⁻¹, the storage modulus value (G') of the Gel-H- Ca^{2+} was as high as 65.4 kPa, demonstrating a tough performance. However, the loss tangent value ($\tan \delta$) of the Gel-H- Ca^{2+} was ≈ 0.35 , higher than that of the Gels-L- Ca^{2+} (≈ 0.14). This higher energy loss suggested that the cross-linking structure between Ca^{2+} ions and the hydroxyl group was unstable and easily damaged by external forces. To investigate the visco-elastic transition, the strain sweep test was performed between 0.01% and 100% at constant frequency. As the concentration of Ca^{2+} ion increased, the visco-elastic transition point increased from 2% to 10% (Figure S2, Supporting Information), indicating that high concentration of Ca^{2+} ions effectively improved the strength and stability of the cross-linking structure, which required higher strains to destroy the original gel structure.

Cyclic compression loading–unloading test was used to explore the energy dissipation mechanism of cellulose hydrogels. From the curves in Figure 2d, the mechanical hysteresis was clearly observed for two hydrogels. At the same 20% strain, the hysteresis areas of the Gel-L- Ca^{2+} and Gel-H- Ca^{2+} were 93.1 and 136.4 KJ m⁻³, respectively. It suggests that ionic cross-linking are reversible coordination bonds that can dissipate energy in these hydrogels. A higher cross-link density could lead to a higher energy loss in the hysteresis cycle. After 30 min of waiting time, the recovery area ratio of Gel-L- Ca^{2+} is 75%, while that of Gel-H- Ca^{2+} is 54%. The lower ratio of the Gel-H- Ca^{2+} indicates the hydrogel is difficult to recover after compression due to its high cross-linking density.

In addition, zero stress at initial compression could be observed in the second loading–unloading cycle. It indicated that the hydrogel underwent an irreversible shape change after the first compression. Therefore, a continuous compress upon the hydrogel was further conducted for a longer time to investigate the compression effects on hydrogel structure. It can be seen from Figure 2e that the stress was greatly reduced when the compress time was prolonged, which indicated that the original cross-linking was rapidly destroyed. To show the effect more clearly, a cube Gel-H- Ca^{2+} ($1 \times 1 \times 1$ cm³) was compressed to 60% of its initial width and held for 6 h. The hydrogel could maintain the compressed shape and not rebound when the pressure was relieved (Figure 2f). It also revealed the cross-linking structure is easy to be reconstructed under an external force.

For cross-sectional observation by SEM, the samples were first deionized, and then transferred to liquid nitrogen to fracture. Although they are different from the original state, the microstructure can still be observed from SEM images as shown in Figure 2g–j. Sol-H- Zn^{2+} had no obvious structural features as expected. However, Gel-L- Ca^{2+} showed an ordered morphology, indicating that the alignment of cellulose chains occurred in $\text{Ca}^{2+}/\text{Zn}^{2+}$ ion exchange process. This may be due to the fact

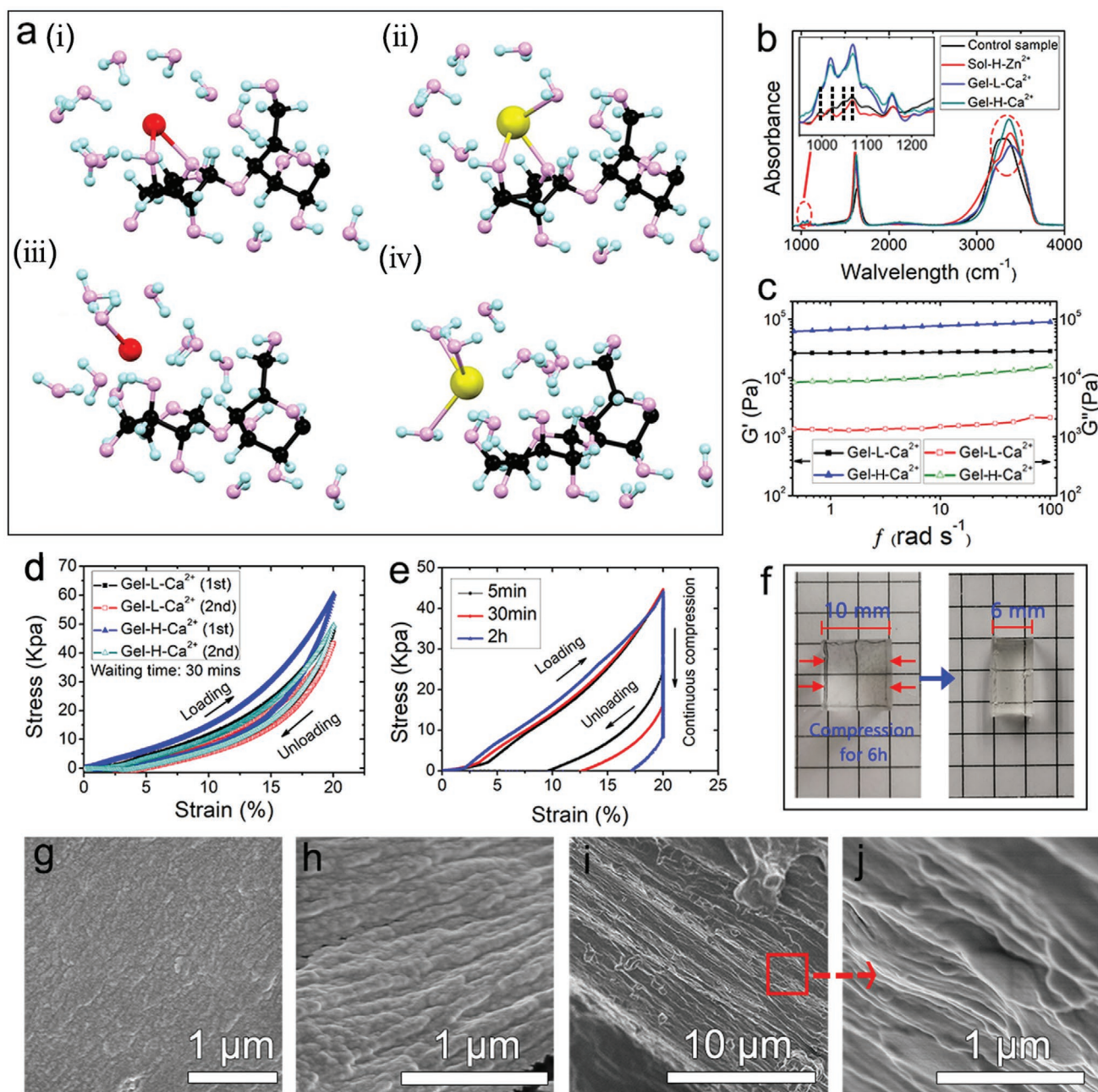


Figure 2. a) Favorable coordination configurations of interactions between metal ions and oxygen from sugar ring and hydroxyl groups of cellulose chains in water media (red ball: Zn^{2+} , yellow ball: Ca^{2+}), b) FT-IR spectra of various cellulose hydrogel with spectra in the region of $950\text{--}1250\text{cm}^{-1}$ as inset, c) dynamic viscoelasticity of the cellulose hydrogels with storage modulus (G') and loss modulus (G'') as a function of frequency, d) cyclic loading–unloading curves of the Gel-L- Ca^{2+} and Gel-H- Ca^{2+} with a waiting time of 30 min for each repeat, e) the stress–strain curves of Gel-H- Ca^{2+} under 20% strain with a continuous compression for different time, f) digital images of Gel-H- Ca^{2+} compressed to 60% of its thickness, and SEM images of g) Sol-H- Zn^{2+} , h) Gel-L- Ca^{2+} , i,j) Gel-H- Ca^{2+} .

that cellulose is a relatively rigid polymer and the straight chains are more easily oriented in the cross-linking process. With the increase of Ca^{2+} ion content, Gel-H- Ca^{2+} exhibited fibrous structure and a significant anisotropic structure, which further confirmed the formation of a uniform cross-linking network. Therefore, Gel-H- Ca^{2+} could show a bright interference color with slight tension or compression under cross-polarized light. To further verify the effect of cellulose chain orientation on the mechanical

properties of the gel, we conducted the compressive property test of pre-compressed Gel-H- Ca^{2+} samples along parallel and perpendicular to the cellulose alignment direction. The fracture stress and strain of Gel-H- Ca^{2+} when the chain orientation is parallel to the compressive direction (Figure S3, Supporting Information) were 2.7 MPa and 62%, significantly higher than that of perpendicular direction (1.7 MPa, 52%), indicating that the anisotropic structure can affect the mechanical properties of cellulose gel.

In order to verify the dynamic orientation transition of the molecular chain, an in situ $\text{Ca}^{2+}/\text{Zn}^{2+}$ ion exchange process was designed and observed. Each hydrogel sample was placed on the sample stage and fixed at 45° under cross-polarization (Figure 3a,b). As shown in Figure 3c, the initial interference color of Gel-L- Ca^{2+} is uniformly light yellow (first color order). The interference order was determined by the classical Michel-Levy interference color chart. When the hydrogel was soaked in a high concentration of CaCl_2 solution (5.77 mol L^{-1}) for 15 min, the large area of red (first order) appeared in the middle area of the hydrogel, surrounded by blue-green (second color order). Considering that there was no change in the size of the hydrogel during the ion exchange process, the increase of interference color order should come from the cross-linking by Ca^{2+} ions. As the ions diffused from the edge to the center, a color gradient appeared. When the soaking time of Ca^{2+} ions solution extended to 125 min, the green region in the middle gradually became pink (second color order), suggesting that the retardation continued to increase. At the same time, the color gradient was more obvious, indicating an increased ordered structure in overall hydrogel.

Starting from 125 min, the gel was soaked in a low concentration ZnCl_2 solution (4.21 mol L^{-1}). At 140 min, the distinguished contour lines began to blur and the middle part of the hydrogel turned to yellow. When the time extended to 155 min, the remaining red in the middle region disappeared and then completely turned to yellow. During this process, the interference order reduced by one order. When the soaking time extended to 275 min, the yellow in the middle region gradually turned to blue, and the interference color order continued to decrease by one order. In the course of 275–335 min, the blue region slowly changed to magenta, and finally returned to the original color. From the above phenomena, it can be determined that the cross-linking degree of cellulose chains can be effectively reduced by the coordination of Zn^{2+} . Importantly, the anisotropic regulation of cellulose hydrogels by $\text{Ca}^{2+}/\text{Zn}^{2+}$ ion exchange is reversible and can be repeated for several times. It should be overemphasized that this is the first report that the anisotropy of hydrogels could be controlled by ion exchange process.

The birefringence (Δn) of the sample was calculated from the measurement of crossed transmittance ($T_\perp$) and parallel transmittance ($T_\parallel$) curves using the method reported in the literature (details in Supporting Information).^[21] The transmittance test method of the hydrogels in crossed and parallel polarization was specifically depicted in Figure 3d. The crossed and parallel transmittance spectra of the hydrogels changed significantly, accompanied with the ion exchange process. As expected, the $T_\perp$ spectrum went through a cycle from 0 to 320 min and finally returned to its original state, which was well matched to the change of interference color. According to calculations and Cauchy formulas, the birefringence at different wavelengths was obtained by least squares fitting. During the Ca^{2+} ion substituting process, the birefringence increases from 0.5×10^{-4} to 2.2×10^{-4} (Figure 3e). In contrast, during the Zn^{2+} ion exchange process, the birefringence decreases to 0.5×10^{-4} , returning to the original value.

Based on optical characteristics of the hydrogels, the optical sensitivity of Gel-H- Ca^{2+} was further explored under a slight

external compressive force. The micro-controlled compress deformation setup was depicted in Figure 4a. During the dynamic monitoring process with a deformation interval of 1% to its original thickness, the main peak of transmittance spectra under the crossed polarization gradually red shifted (Figure 4b), indicating a good optical sensing property to pressure response. In another experiment, the deformation interval of 4% was applied. When the deformation ratio increases from 0% to 16%, the peak red shifted throughout the visible range (Figure S4, Supporting Information). Furthermore, when the deformation ratio increased to 20%, the peak moved back to the low wavelength band and red-shifted again with the deformation continued to increase. It indicated that the large deformation could cause the interference color to rise by one order.

A pressure sensor system was built to monitor human activity. Here Gel-H- Ca^{2+} sample was fixed to a finger by a rubber band (Figure 4c). When the finger bent with a small degree, the interference color under crossed polarization changed from purple to blue and then yellow. As the degree of bending of the finger increased, the gel showed a richer color due to uneven forces (the change of interference color when fingers are bent to different degrees was recorded in Video S2, Supporting Information). Considering the conductivity of ionic hydrogel, a simple strain sensor was assembled by connecting the Gel-H- Ca^{2+} to the test circuit and the relative resistance variation ($\Delta R/R_0 = [R - R_0]/R_0$, where R and R_0 correspond to the resistance corresponding to different bending states) as a function of tensile strain was recorded. The $\Delta R/R_0$ increased as the bending degree of the finger increased (details in Figure S5, Supporting Information). When the finger underwent rapid bending–releasing for several cycles, the $\Delta R/R_0$ changed synchronized perfectly with the finger motions (Figure 4d), revealing its instantaneity and good stability. The reason for the steady incline in resistance seen in Figure 4d was due to dissipation of energy caused by breaking of ionic cross-linking from the deformation of the gel. In addition, Gel-H- Ca^{2+} was assembled into a capacitor. The capacitance changed reversibly between 389 and 566 pF in response to the pressing and releasing motion without any hysteresis (Figure 4e). The high optical and electrical sensing properties of the hydrogel under pressure provide a novel idea for the design of versatile sensors and multi-modal bionics.

3. Conclusion

In this work, three state-switchable cellulose hydrogels were designed and fabricated through $\text{Ca}^{2+}/\text{Zn}^{2+}$ ion exchange at room temperature. The microstructure transition in such process could be observed by the change of the interference color. Introduction of Ca^{2+} ions leads to formation of cellulose hydrogel with enhanced compressive strength and higher orientation, while the incorporation of Zn^{2+} ions leads to fluidity and low orientation. The cellulose ionic hydrogels with a high Ca^{2+} content exhibited good optical and electrical sensing properties to pressure response, which provides novel ideas and designs for the construction of multifaceted sensors and bionics.

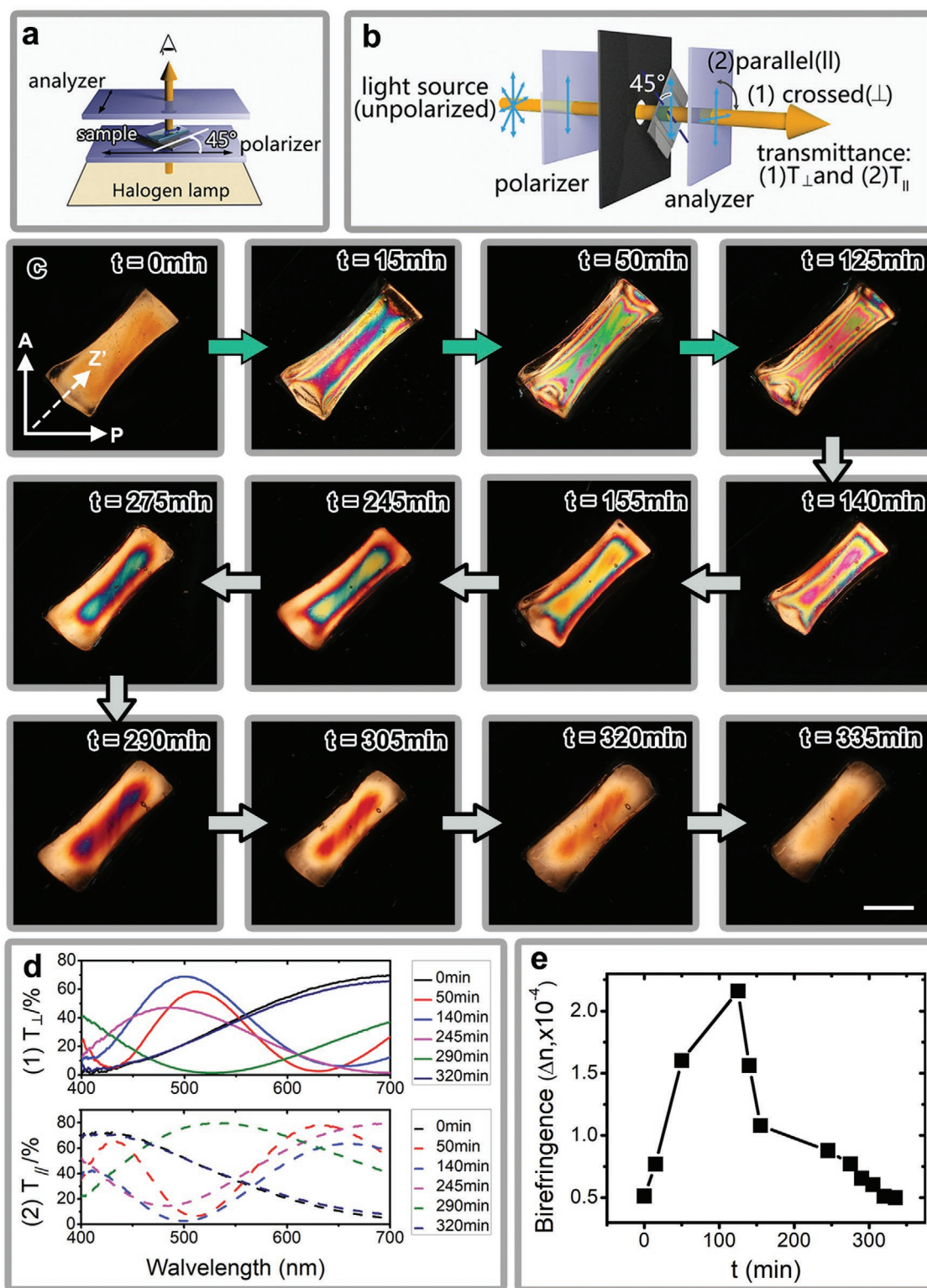


Figure 3. a) Schematic diagram of the sample and the observation direction, the polarization axis was settled at 45° to the compression axis of the cellulose gel, b) experimental setup for measuring the transmittance of ion exchanged hydrogel under crossed ($T_{\perp}$) and parallel ($T_{\parallel}$) polarization, c) images of ion exchange process under cross-polarization (scale bar = 10 mm), d) transmittance spectra of hydrogels under crossed and parallel polarization, e) the birefringence at 500 nm of the hydrogel estimated by fitting the Cauchy formula.

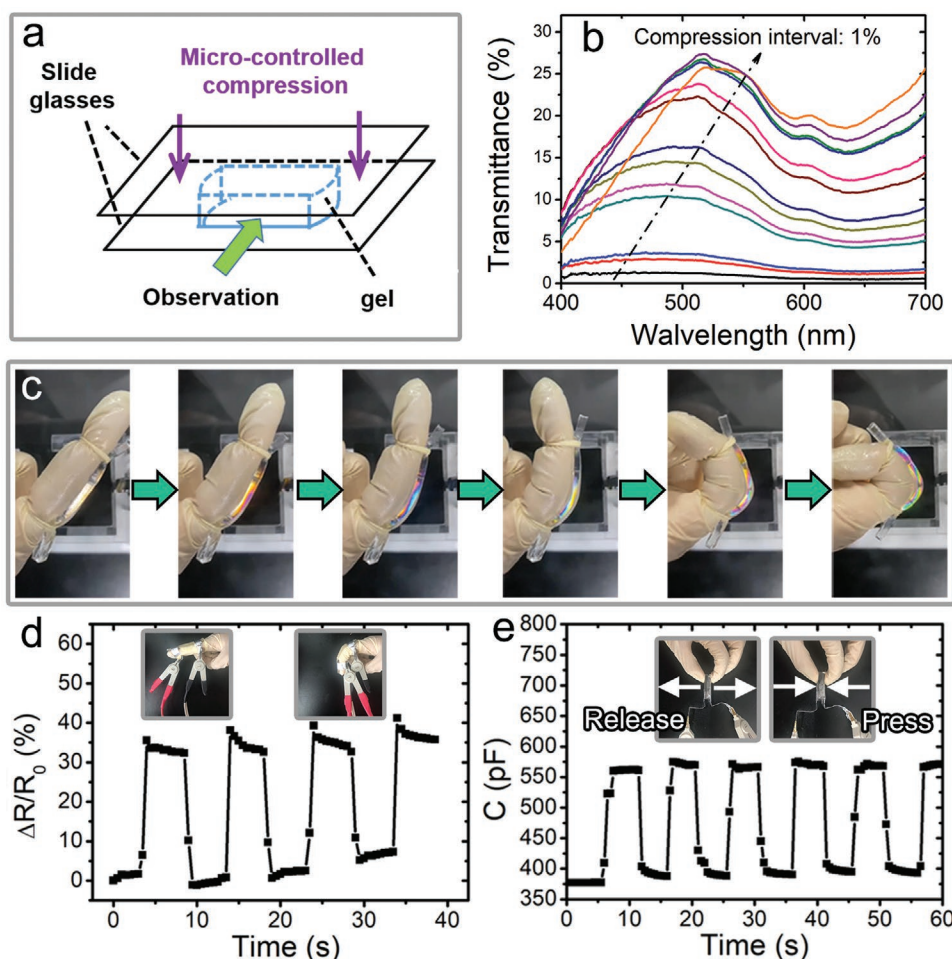


Figure 4. a) Schematic diagram of the micro-controlled compress deformation upon the Gel-H- Ca^{2+} , b) trends of spectral changes with a compression ratio interval of 1% under crossed polarization, c) optical sensing of cellulose gel induced by different degree of bending of a finger under crossed polarization, d) cyclic resistance changes with finger straightening and bending, and e) cyclic capacitance change under finger press-release.

4. Experimental Section

Materials: Cellulose (cotton linter pulp) with α -cellulose content exceeding 96% was provided by Nantong Cellulose Fiber Company. Zinc chloride (ZnCl_2 , 99.0%) and calcium chloride (CaCl_2 , 99.0%) were purchased from Nanjing Reagent Company. All reagents were used as received unless otherwise stated.

Preparation of Cellulose Hydrogel: 9.87 g of ZnCl_2 and 0.3 g of CaCl_2 were dissolved in 3.63 g of deionized water at 75 °C. Then, 0.2 g of cotton linter was added to above ionic compounds and stirred vigorously for 30 min. After the cotton linter was completely dissolved, 2.5 g of deionized water was added, followed with sonication to remove the bubbles. The mixture was poured into a set mold at room temperature to obtain the cellulose hydrogel.

Three-State Reversible Transformation: The cellulose hydrogel was soaked in a 5.57 mol L^{-1} CaCl_2 solution for 2 h and repeated for three times until most of Zn^{2+} ions were exchanged to obtain a hydrogel with high Ca^{2+} content (Gel-H- Ca^{2+}). Then the hydrogel was soaked in a 4.21 mol L^{-1} ZnCl_2 solution for 12 h. After most of the Ca^{2+} were exchanged by the Zn^{2+} , a brittle hydrogel (Gel-L- Ca^{2+}) was obtained. After the brittle hydrogel was soaked in a 7.98 mol L^{-1} ZnCl_2 solution for 12 h, the gel was dissolved into a fluid (Sol-H- Zn^{2+}). For a reverse process, the fluid was soaked in a 4.33 mol L^{-1} CaCl_2 solution for 30 min, a brittle hydrogel was obtained. The brittle solid gel was then

soaked in a 5.57 mol L^{-1} CaCl_2 solution for 12 h and turned into the rigid hydrogel. The entire ion exchange process was conducted at room temperature.

Additional Measurements and Characterization: The microstructure of the hydrogel was characterized by field emission scanning electron microscopy (FSEM, Model-S4800). To prepare FSEM samples, the gel was soaked in tert-butanol, and then transferred to liquid nitrogen to make it brittle to fracture to observe the cross section of the sample. Fourier transform infrared (FT-IR) spectra were recorded on a Bruker (VERTEX 80 V) to characterize interaction between metal ions and hydroxyl groups during ion exchange, and the spectral range was 500–4000 cm^{-1} . Flame atomic absorption measurements were carried out using a PerkinElmer Model AA900T atomic absorption spectrometer to determine the ion concentration during ion exchange, operating with an air-acetylene flame that had previously been calibrated using calcium and zinc standard solutions. The instrument was set at 422.7 nm for calcium and 213.9 nm for zinc, respectively.

The compression test of the gel was performed at room temperature using a commercial tester (CMT850). A cylindrical sample with a thickness of 15 mm and a diameter of 22 mm was used. The compression rate was 10 mm min^{-1} . The rheological behavior of gels (22.0 mm in diameter and 2.0 mm in thickness) was conducted by Rheowin MAS60. The frequency sweeps were performed over a range of 0.01–100 rad s^{-1} at a controlled strain amplitude of 1% at room temperature.

The orientation transition test was performed. The cellulose hydrogel ($2 \times 1 \times 1 \text{ cm}^3$) was compressed into 0.8 cm along the thickness direction and kept for 6 h to ensure the compressed shape was fixed. Each hydrogel sample was placed on the sample stage and fixed at 45° under cross-polarization. Field of view was perpendicular to the compression axis. The compressed hydrogel was then soaked in 30 mL of $4.21 \text{ mol L}^{-1} \text{ ZnCl}_2$ solution for 210 min and such hydrogel was set as the initial state. The hydrogel was soaked in 30 mL of $5.57 \text{ mol L}^{-1} \text{ CaCl}_2$ solution for different time. In this process, digital photos under crossed polarization were taken to in situ observe the interference color change. UV-vis spectroscopy analysis was carried out on an Ocean Optics 2000+ UV-Vis/NIR spectrophotometer in conjunction with a polarizing microscope.

The experiments involving human subjects were performed with the full, informed consent of the volunteers in accordance with the local laws.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

cellulose, hydrogels, ions exchange, orientation, pressure sensing

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